Year 11 Higher

REVISION PACK

Progress Check 3

Thursday 12 November 2026

Name: _____

Class: _____

Units in this pack

- Number 9
- Algebra 9

Number 9 · Percentages of amounts and percentage change (Lessons 6.1, 6.2)

Worked example: a percentage of an amount

A tank holds 800 litres. It is 6.5% full. How many litres of water are in the tank?

Step 1	Change % to a decimal	$6.5\% = 0.065$
Step 2	Multiply	$0.065 \times 800 = 52$
Step 3	Answer	52 litres

Practice: a percentage of an amount

1.1 A water tank holds 1500 litres. The tank is 8.4% full. How many litres of water are in the tank?

1.2 Ahmed pays 3.5% of 2400 KD as a fee. How much is the fee?

1.3 There are 320 students in a year group. 62.5% of them walk to school. How many students walk to school?

1.4 A pool holds 2500 litres. The pool is 4.8% full. How many litres of water are in the pool?

1.5 Huda pays 12.5% of 640 KD as tax. How much is the tax?

1.6 A bag of rice weighs 5000 grams. 2.4% of the rice is water. How many grams of water are in the bag?

Number 9 · Percentages of amounts and percentage change (Lessons 6.1, 6.2)

Worked example: a percentage of the rest

Ali earns 2000 KD a month. Ali saves 10% of it. Ali spends 25% of the rest on food. How much does Ali spend on food?

Step 1	What is left	$100\% - 10\% = 90\%$
Step 2	Find the rest	$2000 \times 0.9 = 1800$
Step 3	25% of the rest	$1800 \times 0.25 = 450$
Step 4	Answer	450 KD

Practice: a percentage of the rest

1.7 Mona earns 1500 KD a month. Mona saves 20% of it. Mona spends 30% of the rest on rent. How much does Mona spend on rent?

1.8 Omar earns 2400 KD a month. Omar spends 15% of it. Omar spends 30% of the rest on transport. How much does Omar spend on transport?

1.9 Layla earns 1600 KD a month. Layla saves 25% of it. Layla spends 35% of the rest on food. How much does Layla spend on food?

1.10 Hamad earns 2000 KD a month. Hamad saves 30% of it. Hamad spends 45% of the rest on rent. How much does Hamad spend on rent?

1.11 Dana earns 1250 KD a month. Dana spends 8% of it. Dana spends 20% of the rest on food. How much does Dana spend on food?

1.12 Bader earns 3000 KD a month. Bader saves 12% of it. Bader spends 25% of the rest on travel. How much does Bader spend on travel?

Number 9 · Percentages of amounts and percentage change (Lessons 6.1, 6.2)

Worked example: percentage profit

A shop buys a lamp for 20 KD. It sells the lamp for 27 KD. Work out the percentage profit.

Step 1	Profit	$27 - 20 = 7$
Step 2	Divide by the cost	$7 \div 20 = 0.35$
Step 3	Multiply by 100	$0.35 \times 100 = 35$
Step 4	Answer	35%

Practice: percentage profit

1.13 A shop buys a clock for 30 KD. It sells it for 37.5 KD. Work out the percentage profit.

1.14 A shop buys a mirror for 16 KD. It sells it for 21.6 KD. Work out the percentage profit.

1.15 A shop buys a used bike for 250 KD. It sells it for 210 KD. Work out the percentage loss.

1.16 A shop buys a kettle for 45 KD. It sells it for 54 KD. Work out the percentage profit.

1.17 A shop buys a fan for 60 KD. It sells it for 75.6 KD. Work out the percentage profit.

1.18 A shop buys a used phone for 320 KD. It sells it for 272 KD. Work out the percentage loss.

Number 9 · Compound interest (Lesson 6.3)

Worked example: compound interest

Sara invests 3000 KD for 4 years. The investment earns compound interest of 2% per year. Work out the value at the end of 4 years. Give your answer correct to the nearest KD.

Step 1	Multiplier	$100\% + 2\% = 102\% = 1.02$
Step 2	Power = years	3000×1.02^4
Step 3	Calculate	$3247.296\ldots$
Step 4	Answer (nearest KD)	$3247\ \text{KD}$

Practice: compound interest

2.1 Yousef invests 5000 KD for 3 years. The investment earns compound interest of 3% per year. Work out the value of the investment at the end of 3 years. Give your answer correct to the nearest KD.

2.2 Mariam invests 2800 KD for 4 years. The investment earns compound interest of 1.5% per year. Work out the value of the investment at the end of 4 years. Give your answer correct to the nearest KD.

2.3 Faisal invests 12 000 KD for 5 years. The investment earns compound interest of 4.5% per year. Work out the value of the investment at the end of 5 years. Give your answer correct to the nearest KD.

2.4 Noura invests 6000 KD for 5 years. The investment earns compound interest of 2% per year. Work out the value of the investment at the end of 5 years. Give your answer correct to the nearest KD.

2.5 Ahmed invests 1800 KD for 2 years. The investment earns compound interest of 3.5% per year. Work out the value of the investment at the end of 2 years. Give your answer correct to the nearest KD.

2.6 Reem invests 9500 KD for 3 years. The investment earns compound interest of 1.2% per year. Work out the value of the investment at the end of 3 years. Give your answer correct to the nearest KD.

Number 9 · Compound interest (Lesson 6.3)

Worked example: after how many years?

The population of a town is 40 000. It decreases by 5% each year. After how many whole years will it first be less than 35 000?

Step 1	Multiplier	$100\% - 5\% = 95\% = 0.95$
Step 2	Try 2 years	$40000 \times 0.95^2 = 36100$ (not yet)
Step 3	Try 3 years	$40000 \times 0.95^3 = 34295$ (< 35000)
Step 4	Answer	3 years

Practice: after how many years?

2.7 The population of a town is 50 000. The population decreases by 6% each year. After how many whole years will the population first be less than 40 000?

2.8 A machine is worth 8000 KD. Its value falls by 15% each year. After how many whole years will it first be worth less than 4000 KD?

2.9 Salem invests 2000 KD at 8% compound interest per year. After how many whole years will the investment first be worth more than 3000 KD?

2.10 The population of a city is 30 000. The population decreases by 3% each year. After how many whole years will the population first be less than 25 000?

2.11 A van is worth 12 000 KD. Its value falls by 10% each year. After how many whole years will it first be worth less than 7000 KD?

2.12 Amal invests 5000 KD at 4% compound interest per year. After how many whole years will the investment first be worth more than 6000 KD?

Number 9 · Reverse percentages (Lesson 6.4)

Worked example: price before a reduction

In a sale, all prices are reduced by 20%. The sale price of a coat is 64 KD. Work out the price of the coat before the sale.

Step 1	Sale price is 80%	$100\% - 20\% = 80\% = 0.8$
Step 2	Divide by 0.8	$64 \div 0.8 = 80$
Step 3	Check	$80 \times 0.8 = 64$✓
Step 4	Answer	$80\ \text{KD}$

Practice: price before a reduction

3.1 In a sale, all prices are reduced by 15%. The sale price of a coat is 85 KD. Work out the price of the coat before the sale.

3.2 In a sale, all prices are reduced by 30%. The sale price of a pair of shoes is 42 KD. Work out the price before the sale.

3.3 The value of a computer fell by 8%. It is now worth 1748 KD. Work out its value before the fall.

3.4 In a sale, all prices are reduced by 25%. The sale price of a table is 120 KD. Work out the price before the sale.

3.5 The price of a phone is reduced by 10% to 76.5 KD. Work out the price before the reduction.

3.6 The value of a car fell by 5%. It is now worth 2850 KD. Work out its value before the fall.

Number 9 · Reverse percentages (Lesson 6.4)

Worked example: how much more after a rise?

After a pay rise of 10%, Ali earns 1320 KD a month. How much more does Ali earn each month after the pay rise?

Step 1	New pay is 110%	$100\% + 10\% = 110\% = 1.1$
Step 2	Old pay	$1320 \div 1.1 = 1200$
Step 3	Subtract	$1320 - 1200 = 120$
Step 4	Answer	120 KD

Practice: how much more after a rise?

3.7 After a pay rise of 12%, Salem earns 1680 KD a month. How much more does Salem earn each month after the pay rise?

3.8 After a pay rise of 5%, Noura earns 2016 KD a month. How much more does Noura earn each month after the pay rise?

3.9 After a pay rise of 2%, Hamad earns 943.5 KD a month. How much more does Hamad earn each month after the pay rise?

3.10 After a pay rise of 10%, Mariam earns 1265 KD a month. How much more does Mariam earn each month after the pay rise?

3.11 After a pay rise of 4%, Yousef earns 2340 KD a month. How much more does Yousef earn each month after the pay rise?

3.12 After a pay rise of 5%, Omar earns 1428 KD a month. How much more does Omar earn each month after the pay rise?

Algebra 9 · Direct and inverse proportion (Lessons 7.1, 7.2)

Worked example: finding the formula, then using it backwards

m is directly proportional to the cube of r. When $r = 2$, $m = 40$

(a) Find a formula for m in terms of r. (b) Work out the value of r when $m = 135$

(a)	Step 1	Write with k	$m = kr^3$
	Step 2	Put in $r = 2$, $m = 40$	$40 = k \times 8$
	Step 3	Find k	$k = 5$
	Step 4	Answer	$m = 5r^3$
(b)	Step 1	Put in $m = 135$	$135 = 5r^3$
	Step 2	Divide by 5	$r^3 = 27$
	Step 3	Cube root	$r = 3$
	Step 4	Answer	$r = 3$

Practice: finding the formula, then using it backwards

4.1 m is directly proportional to the cube of r. When $r = 2$, $m = 24$

(a) Find a formula for m in terms of r. (b) Work out the value of r when $m = 81$

4.2 m is directly proportional to the cube of r. When $r = 3$, $m = 54$

(a) Find a formula for m in terms of r. (b) Work out the value of r when $m = 128$

4.3 m is directly proportional to the square of r. When $r = 4$, $m = 80$

(a) Find a formula for m in terms of r. (b) Work out the value of r when $m = 125$

4.4 m is directly proportional to the cube of r. When $r = 2$, $m = 16$

(a) Find a formula for m in terms of r. (b) Work out the value of r when $m = 250$

4.5 m is directly proportional to the cube of r. When $r = 4$, $m = 320$

(a) Find a formula for m in terms of r. (b) Work out the value of r when $m = 135$

4.6 m is directly proportional to the square of r. When $r = 3$, $m = 36$

(a) Find a formula for m in terms of r. (b) Work out the value of r when $m = 100$

Algebra 9 · Direct and inverse proportion (Lessons 7.1, 7.2)

Worked example: inverse proportion to x²

y is inversely proportional to x^2. When $x = 2$, $y = 9$. Work out the value of y when $x = 3$

Step 1	Write with k	$y = \frac{k}{x^2}$
Step 2	Put in $x = 2$, $y = 9$	$9 = \frac{k}{4}$
Step 3	Find k	$k = 36$
Step 4	Put in $x = 3$	$y = \frac{36}{9}$
Step 5	Answer	$y = 4$

Practice: inverse proportion to x²

4.7 y is inversely proportional to x^2. When $x = 3$, $y = 8$ Work out the value of y when $x = 6$

4.8 y is inversely proportional to x^2. When $x = 2$, $y = 20$ Work out the value of y when $x = 4$

4.9 y is inversely proportional to x^2. When $x = 5$, $y = 4$ Work out the value of y when $x = 2$

4.10 y is inversely proportional to x^2. When $x = 4$, $y = 5$ Work out the value of y when $x = 2$

4.11 y is inversely proportional to x^2. When $x = 1$, $y = 36$ Work out the value of y when $x = 3$

4.12 y is inversely proportional to x^2. When $x = 10$, $y = 1.5$ Work out the value of y when $x = 5$

Algebra 9 · Direct and inverse proportion (Lessons 7.1, 7.2)

Worked example: direct proportion to √x

y is directly proportional to $\sqrt{x}$. When $x = 16$, $y = 12$. Work out the value of y when $x = 25$

Step 1	Write with k	$y = k\sqrt{x}$
Step 2	Put in $x = 16$, $y = 12$	$12 = k \times 4$
Step 3	Find k	$k = 3$
Step 4	Put in $x = 25$	$y = 3 \times 5$
Step 5	Answer	$y = 15$

Practice: direct proportion to √x

4.13 y is directly proportional to $\sqrt{x}$. When $x = 4$, $y = 10$ Work out the value of y when $x = 36$

4.14 y is directly proportional to $\sqrt{x}$. When $x = 25$, $y = 15$ Work out the value of y when $x = 64$

4.15 y is directly proportional to $\sqrt{x}$. When $x = 9$, $y = 6$ Work out the value of y when $x = 81$

4.16 y is directly proportional to $\sqrt{x}$. When $x = 16$, $y = 8$ Work out the value of y when $x = 100$

4.17 y is directly proportional to $\sqrt{x}$. When $x = 4$, $y = 14$ Work out the value of y when $x = 9$

4.18 y is directly proportional to $\sqrt{x}$. When $x = 36$, $y = 30$ Work out the value of y when $x = 4$

Algebra 9 · Direct and inverse proportion (Lessons 7.1, 7.2)

Worked example: what happens to y?

y is inversely proportional to x^2. x is doubled. Write down what happens to y.

Step 1	x doubled, so x^2	$2^2 = 4$
Step 2	Inverse: do the opposite	$\div 4$
Step 3	Answer	y is divided by 4

Practice: what happens to y?

4.19 y is directly proportional to x^2. x is multiplied by 3. Write down what happens to y.

4.20 y is inversely proportional to x^2. x is multiplied by 2. Write down what happens to y.

4.21 y is directly proportional to x^3. x is multiplied by 2. Write down what happens to y.

4.22 y is inversely proportional to x. x is multiplied by 5. Write down what happens to y.

4.23 y is directly proportional to $\sqrt{x}$. x is multiplied by 4. Write down what happens to y.

4.24 y is inversely proportional to x^2. x is multiplied by 10. Write down what happens to y.